

SAT Math Mastery · Week 8A

Angles, Similarity & Right-Triangle Trig

Time 150 minutes: 110 min solving + 40 min self-check and error log

Topics parallel lines & angles · triangle sum, exterior angle & triangle inequality · similar triangles & scaling of lengths/areas · Pythagorean triples & special right triangles · sine, cosine, tangent · $\sin x^\circ = \cos(90 - x)^\circ$ · triangle area

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Use vertical, linear-pair, corresponding, alternate-interior and co-interior angle relations, including with algebraic angle measures.
- S2.** Use the triangle angle sum, the exterior-angle theorem, and the triangle inequality (which side lengths can or cannot form a triangle).
- S3.** Prove or use similarity (AA, parallel line cutting a triangle) and set up proportions between corresponding sides, altitudes and perimeters.
- S4.** Scale lengths by k , areas by k^2 (and volumes by k^3); convert between a scale ratio and an area ratio or a percent change.
- S5.** Apply the Pythagorean theorem, Pythagorean triples, and the 45° - 45° - 90° and 30° - 60° - 90° side ratios.
- S6.** Write $\sin$, $\cos$, $\tan$ as ratios of sides; solve for a side from an angle (calculator in degree mode).
- S7.** Use the complementary-angle identity $\sin x^\circ = \cos(90 - x)^\circ$ and the fact that trig ratios depend only on the angle, not on the size of the triangle.
- S8.** Compute triangle area from base and height or from two sides and the included angle, and turn area information into a missing length or angle.

2 · REQUIRED FACTS

- Two intersecting lines make vertical angles (equal) and linear pairs (sum 180°). Parallel lines cut by a transversal give eight angles, each equal to θ or to $180^\circ - \theta$.
- Corresponding and alternate-interior angles are equal; co-interior (same-side interior) angles are supplementary.
- The angles of a triangle sum to 180° ; an exterior angle equals the sum of the two remote interior angles. The angles of an n -gon sum to $(n - 2)180^\circ$.
- Triangle inequality: each side is less than the sum of the other two, and greater than their difference. Equality means the triangle is degenerate (a flat segment).
- A line parallel to one side of a triangle cuts the other two sides proportionally and creates a smaller triangle similar to the whole.
- Similarity ratio k : lengths (sides, altitudes, perimeters) scale by k , areas by k^2 . Angles never change.
- Common triples: 3-4-5, 5-12-13, 8-15-17, 7-24-25, 9-40-41, and their multiples.
- In a right triangle the two acute angles are complementary, so $\sin A = \cos B$ and $\tan A \cdot \tan B = 1$ when $A + B = 90^\circ$.
- The altitude to the hypotenuse of a right triangle splits it into two triangles similar to the whole triangle and to each other.

3 · REQUIRED FORMULAS

$$\angle_{\text{ext}} = \angle_1 + \angle_2, \quad |a - b| < c < a + b, \quad a^2 + b^2 = c^2, \quad \frac{\text{new}}{\text{old}} = k \Rightarrow \frac{\text{area}_{\text{new}}}{\text{area}_{\text{old}}} = k^2$$

$$45^\circ\text{-}45^\circ\text{-}90^\circ : s, s, s\sqrt{2} \quad 30^\circ\text{-}60^\circ\text{-}90^\circ : s, s\sqrt{3}, 2s \quad \text{equilateral: } h = \frac{\sqrt{3}}{2}s, A = \frac{\sqrt{3}}{4}s^2$$

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}, \quad \cos \theta = \frac{\text{adj}}{\text{hyp}}, \quad \tan \theta = \frac{\text{opp}}{\text{adj}}, \quad \sin x^\circ = \cos(90 - x)^\circ, \quad A = \frac{1}{2}bh = \frac{1}{2}ab \sin C$$

4 · MINI THEORY SHEET

Angle chasing in a figure. Mark every angle you can find in this order: (1) vertical angles and linear pairs, (2) parallel-line relations, (3) triangle sum / exterior angle. When angles carry algebra, decide first whether the two angles are *equal* or *supplementary*, then write one equation.

Which angle relation?

Position of the two angles	Relation
opposite each other at one crossing	equal (vertical)
same corner at two crossings (same side of the transversal)	equal (corresponding)
between the lines, opposite sides of the transversal	equal (alternate interior)
between the lines, same side of the transversal	sum 180° (co-interior)

Similar triangles. Match vertices by angle, write the proportion with matching sides in the same order, then cross-multiply. A segment parallel to a side gives $\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}$ (part-to-whole), and also $\frac{AD}{DB} = \frac{AE}{EC}$ (part-to-part). Do not mix part-to-part with part-to-whole.

Scaling. If lengths scale by k , areas by k^2 . A “ $p\%$ larger area” means a factor $1 + \frac{p}{100}$ on area, so $k = \sqrt{1 + \frac{p}{100}}$.

Right triangles. First look for a triple or a special triangle before squaring anything. A leg is $\sqrt{c^2 - a^2}$; a diagonal of a rectangle is $\sqrt{\ell^2 + w^2}$.

Trigonometry. Label opposite, adjacent and hypotenuse *relative to the angle you are given*. Ratios are properties of the angle, so $\sin A$ is the same in every triangle similar to the one shown. If $\sin A = \cos B$ for acute angles A and B , then $A + B = 90^\circ$.

Triangle area. With two sides a, b and included angle C , $A = \frac{1}{2}ab \sin C$. Since $\sin C = \sin(180^\circ - C)$, an area condition often has *two* possible angles C and $180^\circ - C$.

Speed shortcuts.

Situation	Shortcut
third side of a triangle is a whole number	$ a - b < c < a + b$: count integers strictly between
angle expressions in x , same side between parallels	set the sum to 180° (not equal)
area ratio given, find a length	take the square root of the area ratio first
parallel segment, find a whole side	use $\frac{\text{part}}{\text{whole}}$ with the whole = part + rest
$\sin(\cdot) = \cos(\cdot)$ with unknowns	add the two angle expressions and set the sum to 90
leg and hypotenuse differ by 1	$a^2 = 2b + 1$, so $b = \frac{a^2 - 1}{2}$ when $c = b + 1$
right-triangle ratios in a bigger figure	find the similar small right triangle; the ratio is the same
$\tan A$ given	draw the triangle with legs opp, adj; hypotenuse by triple or Pythagoras

5 · GUIDED EXAMPLES

Example 1 — Parallel lines with algebraic angles

Lines $\ell \parallel m$ are cut by a transversal. Two angles lie *between* the lines on the *same side* of the transversal and measure $(2x + 15)^\circ$ and $(3x - 10)^\circ$. Find both angles.

Same-side interior angles are supplementary: $(2x + 15) + (3x - 10) = 180 \Rightarrow 5x + 5 = 180 \Rightarrow x = 35$.

The angles are $2(35) + 15 = 85^\circ$ and $3(35) - 10 = 95^\circ$. Check: $85 + 95 = 180$.

Example 2 — Similar triangles from a parallel line

In $\triangle ABC$, D lies on $\overline{AB}$ and E on $\overline{AC}$ with $\overline{DE} \parallel \overline{BC}$. If $AD = 4$, $DB = 6$ and $DE = 5$, find BC and the ratio of the areas $[ADE] : [ABC]$.

$\triangle ADE \sim \triangle ABC$ with ratio $k = \frac{AD}{AB} = \frac{4}{4+6} = \frac{2}{5}$ (part over *whole*).

$BC = \frac{DE}{k} = \frac{5}{2/5} = 12.5$. Area ratio $= k^2 = \frac{4}{25}$.

Example 3 — Right-triangle trig and the cofunction identity

(a) In right $\triangle ABC$ with $\angle C = 90^\circ$, $AC = 8$ and $BC = 15$. Find $\sin A$ and $\cos B$.

(b) If $\sin(4x - 2)^\circ = \cos(x + 7)^\circ$ and both angles are acute, find x .

(a) $AB = \sqrt{8^2 + 15^2} = 17$. $\sin A = \frac{BC}{AB} = \frac{15}{17}$, and $\cos B = \frac{BC}{AB} = \frac{15}{17}$, the same value (complementary angles).

(b) Complementary angles: $(4x - 2) + (x + 7) = 90 \Rightarrow 5x + 5 = 90 \Rightarrow x = 17$.

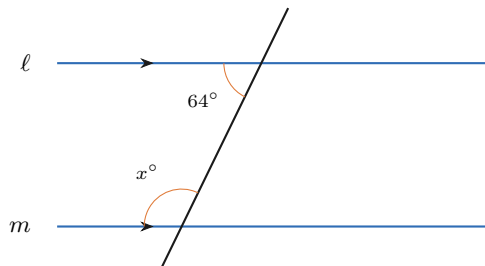
6 · PRACTICE SET — 48 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

EASY · NO CALCULATOR NEEDED

In the figure, line ℓ is parallel to line m . What is the value of x ?



(A) 64

(B) 116 ✓

(C) 26

(D) 154

1. The marked angle at the top and the angle x° lie between the parallel lines on the same side of the transversal (co-interior angles), so they are supplementary.

2. $x = 180 - 64 = 116$.

Answer: (B)

2.

EASY · NO CALCULATOR NEEDED

In triangle ABC , the measure of angle A is 38° . The exterior angle at vertex C , formed by extending side $\overline{BC}$ beyond C , measures 115° . What is the measure, in degrees, of angle B ?

1. An exterior angle equals the sum of the two remote interior angles: $115 = 38 + B$.

2. $B = 115 - 38 = 77$.

Answer: 77

3.

EASY · NO CALCULATOR NEEDED

The lengths of two sides of a triangle are 7 and 12. Which of the following could be the length of the third side?

(A) 20

(B) 5

(C) 10 ✓

(D) 19

1. The third side c must satisfy $12 - 7 < c < 12 + 7$, that is $5 < c < 19$.

2. Only 10 lies strictly between 5 and 19 (5 and 19 would give a flat triangle).

Answer: (C)

4.

EASY · NO CALCULATOR NEEDED

A rectangular banner has a width of 20 inches and a diagonal of 25 inches. What is the area, in square inches, of the banner?

1. The diagonal is the hypotenuse of a right triangle with legs 20 and the length ℓ : $\ell = \sqrt{25^2 - 20^2} = \sqrt{225} = 15$ (a 3-4-5 triple scaled by 5).

2. Area = $20 \cdot 15 = 300$.

Answer: 300

5.

EASY · NO CALCULATOR NEEDED

In right triangle ABC , angle C is the right angle, $AC = 15$ and $BC = 8$. What is the value of $\sin A$?

(A) $\frac{8}{17}$ ✓

(B) $\frac{15}{17}$

(C) $\frac{8}{15}$

(D) $\frac{17}{8}$

1. $AB = \sqrt{15^2 + 8^2} = \sqrt{289} = 17$.

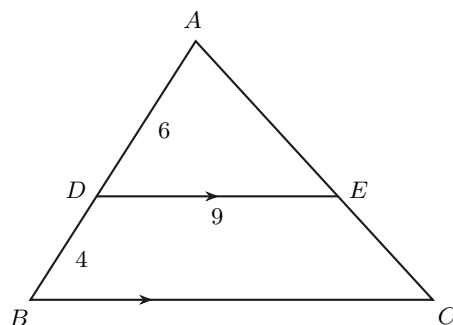
2. The side opposite angle A is $BC = 8$, so $\sin A = \frac{\text{opp}}{\text{hyp}} = \frac{8}{17}$.

Answer: (A)

6.

MEDIUM · NO CALCULATOR NEEDED

In the figure, $\overline{DE}$ is parallel to $\overline{BC}$. What is the length of $\overline{BC}$?



(A) 22.5

(B) 6

(C) 13.5

(D) 15 ✓

1. Since $\overline{DE} \parallel \overline{BC}$, $\triangle ADE \sim \triangle ABC$.

2. $AB = AD + DB = 6 + 4 = 10$, so $\frac{DE}{BC} = \frac{AD}{AB} = \frac{6}{10}$.

3. $BC = \frac{9 \cdot 10}{6} = 15$.

Answer: (D)

7.

MEDIUM · NO CALCULATOR NEEDED

Triangle ABC is similar to triangle DEF , with $AB = 8$ and the corresponding side $DE = 12$. The area of triangle ABC is 40 square units. What is the area, in square units, of triangle DEF ?

1. The scale factor from ABC to DEF is $k = \frac{12}{8} = \frac{3}{2}$.

2. Areas scale by $k^2 = \frac{9}{4}$.

3. Area of $DEF = 40 \cdot \frac{9}{4} = 90$.

Answer: 90

8.

MEDIUM · NO CALCULATOR NEEDED

The measures of the three angles of a triangle are $(3x + 10)^\circ$, $(5x - 14)^\circ$ and $(x + 4)^\circ$. What is the measure, in degrees, of the largest angle?

(A) 116

(B) 70

(C) 86 ✓

(D) 100

1. Angle sum: $(3x + 10) + (5x - 14) + (x + 4) = 180 \Rightarrow 9x = 180 \Rightarrow x = 20$.

2. The angles are 70° , 86° and 24° (sum 180°).

3. The largest is 86° .

Answer: (C)

9.

MEDIUM · NO CALCULATOR NEEDED

The diagonal of a square has length $10\sqrt{2}$. What is the perimeter of the square?

(A) 20

(B) 40 ✓

(C) $40\sqrt{2}$

(D) 100

1. A square's diagonal splits it into two 45° - 45° - 90° triangles, so the diagonal is $s\sqrt{2}$.

2. $s\sqrt{2} = 10\sqrt{2} \Rightarrow s = 10$.

3. Perimeter $= 4s = 40$.

Answer: (B)

10.

MEDIUM · NO CALCULATOR NEEDED

The altitude of an equilateral triangle is $6\sqrt{3}$. What is the perimeter of the triangle?

1. The altitude splits the triangle into two 30° - 60° - 90° triangles; the altitude is the long leg, $\frac{\sqrt{3}}{2}s$.
2. $\frac{\sqrt{3}}{2}s = 6\sqrt{3} \Rightarrow s = 12$.
3. Perimeter = $3 \cdot 12 = 36$.

Answer: 36

11.

MEDIUM · NO CALCULATOR NEEDED

In right triangle ABC , angle C is the right angle and $\tan A = \frac{5}{12}$. What is the value of $\cos B$?

- (A) $\frac{12}{13}$ (B) $\frac{5}{12}$
(C) $\frac{13}{5}$ (D) $\frac{5}{13}$ ✓

1. Take the legs as $BC = 5$ (opposite A) and $AC = 12$ (adjacent to A); then $AB = \sqrt{5^2 + 12^2} = 13$.
2. $\cos B = \frac{\text{adj to } B}{\text{hyp}} = \frac{BC}{AB} = \frac{5}{13}$. (Equivalently $\cos B = \sin A$ since $A + B = 90^\circ$.)

Answer: (D)

12.

MEDIUM · CALCULATOR

A wheelchair ramp makes an angle of 12° with the level ground and has a horizontal run of 15 meters. To the nearest tenth of a meter, how high does the ramp rise?

1. The rise is opposite the 12° angle and the run is adjacent: $\tan 12^\circ = \frac{\text{rise}}{15}$.
2. $\text{rise} = 15 \tan 12^\circ \approx 15(0.2126) \approx 3.19$ (calculator in degree mode).
3. To the nearest tenth: 3.2.

Answer: 3.2

13.

MEDIUM · NO CALCULATOR NEEDED

On a floor plan, 1 centimeter represents 2 meters. A rectangular room has an area of 18 square centimeters on the plan. What is the actual area of the room, in square meters?

(A) 72 ✓

(B) 144

(C) 9

(D) 36

1. Lengths scale by 2 (cm $\rightarrow$ m), so areas scale by $2^2 = 4$.

2. Actual area = $18 \cdot 4 = 72$ square meters.

Answer: (A)

14.

HARD · NO CALCULATOR NEEDED

A triangle has side lengths 8, 15 and x , where $x > 0$. For which of the following values of x is the triangle a right triangle?

I. $x = 17$ II. $x = \sqrt{161}$ III. $x = 23$

(A) II only

(B) I and II only ✓

(C) I, II, and III

(D) I only

1. If x is the hypotenuse: $x^2 = 8^2 + 15^2 = 289 \Rightarrow x = 17$. So I works.

2. If 15 is the hypotenuse: $x^2 = 15^2 - 8^2 = 161 \Rightarrow x = \sqrt{161}$. So II works.

3. $x = 23$ gives $8 + 15 = 23$, which violates the triangle inequality (a flat segment), so III does not form a triangle at all.

4. I and II only.

Answer: (B)

15.

HARD · NO CALCULATOR NEEDED

In triangle ABC , point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that $\overline{DE} \parallel \overline{BC}$. If $AD = x$, $DB = x + 2$, $AE = 6$, $EC = 8$ and $DE = 15$, what is the length of $\overline{BC}$?

1. $\overline{DE} \parallel \overline{BC}$ gives $\frac{AD}{DB} = \frac{AE}{EC}$: $\frac{x}{x+2} = \frac{6}{8} = \frac{3}{4}$.

2. $4x = 3x + 6 \Rightarrow x = 6$, so $AD = 6$, $DB = 8$ and $AB = 14$.

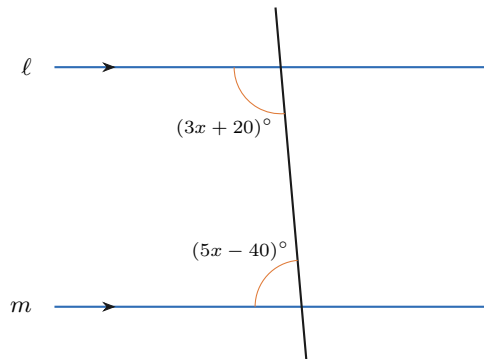
3. $\frac{DE}{BC} = \frac{AD}{AB} \Rightarrow BC = \frac{15 \cdot 14}{6} = 35$.

Answer: 35

16.

HARD · NO CALCULATOR NEEDED

In the figure, line ℓ is parallel to line m . What is the measure, in degrees, of the angle labeled $(5x - 40)^\circ$?



(A) 110

(B) 25

(C) 85 ✓

(D) 95

1. The two labeled angles are between the parallel lines on the same side of the transversal, so they are supplementary.
2. $(3x + 20) + (5x - 40) = 180 \Rightarrow 8x - 20 = 180 \Rightarrow x = 25$.
3. $5x - 40 = 5(25) - 40 = 85$. (Check: the other angle is $3(25) + 20 = 95$, and $85 + 95 = 180$.)

Answer: (C)

17.

HARD · NO CALCULATOR NEEDED

Two parallel lines are cut by a transversal, forming eight angles. One of the eight angles measures 50° . Which of the following could *NOT* be the sum of the measures of some of these eight angles (each angle may be used at most once)?

(A) 500 ✓

(B) 280

(C) 360

(D) 410

1. Each of the eight angles is 50° or 130° : four of each. Any sum has the form $50k + 130m$ with $0 \leq k \leq 4$ and $0 \leq m \leq 4$.
2. $280 = 50(3) + 130(1)$; $360 = 50(2) + 130(2)$; $410 = 50(3) + 130(2)$.
3. 500: for $m = 0, 1, 2, 3, 4$ the remainder $500 - 130m$ is 500, 370, 240, 110, -20. A valid k needs the remainder to equal $50k$ with $0 \leq k \leq 4$ (so at most 200); only 500, 370, 240, 110 are candidates and none is both a multiple of 50 and at most 200. So 500 is impossible.

Answer: (A)

18.

HARD · NO CALCULATOR NEEDED

A triangle has side lengths 6, 11 and k , where k is an integer. If the triangle is *not* isosceles, how many different values of k are possible?

(A) 10

(B) 11

(C) 12

(D) 9 ✓

1. Triangle inequality: $11 - 6 < k < 11 + 6$, so $5 < k < 17$ and $k \in \{6, 7, \dots, 16\}$, which is 11 values.

2. Isosceles happens when $k = 6$ or $k = 11$; remove both: $11 - 2 = 9$.

Answer: (D)

19.

HARD · NO CALCULATOR NEEDED

A right triangle has integer side lengths. Its shorter leg is 11, and its hypotenuse is exactly 1 more than its longer leg. What is the perimeter of the triangle?

1. Let the longer leg be b ; the hypotenuse is $b + 1$. Then $11^2 + b^2 = (b + 1)^2$.

2. $121 + b^2 = b^2 + 2b + 1 \Rightarrow 2b = 120 \Rightarrow b = 60$, hypotenuse 61 (the triple 11-60-61).

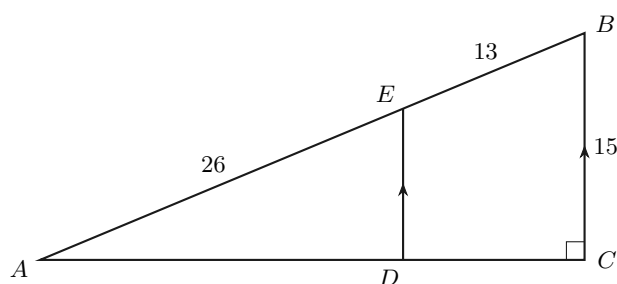
3. Perimeter = $11 + 60 + 61 = 132$.

Answer: 132

20.

HARD · NO CALCULATOR NEEDED

In the figure, $\angle C$ is a right angle and $\overline{DE}$ is parallel to $\overline{BC}$. What is the value of $\cos \angle AED$?



(A) $\frac{5}{12}$

(B) $\frac{2}{3}$

(C) $\frac{5}{13}$ ✓

(D) $\frac{12}{13}$

1. Because $\overline{DE} \parallel \overline{BC}$, $\angle AED = \angle ABC$ (corresponding angles), so $\cos \angle AED = \cos B$.

2. $AB = AE + EB = 26 + 13 = 39$.

3. In right triangle ABC : $\cos B = \frac{BC}{AB} = \frac{15}{39} = \frac{5}{13}$.

Answer: (C)

21.

HARD · NO CALCULATOR NEEDED

If $\sin(2x - 6)^\circ = \cos(3x + 16)^\circ$, where $(2x - 6)^\circ$ and $(3x + 16)^\circ$ are both acute angles, what is the value of x ?

1. Sine and cosine are equal for complementary angles: $(2x - 6) + (3x + 16) = 90$.

2. $5x + 10 = 90 \Rightarrow x = 16$. (Check: 26° and 64° are complementary.)

Answer: 16

22.

HARD · NO CALCULATOR NEEDED

The table gives information about two similar triangular plots of land, X and Y .

Plot	Side lengths (m)	Area (m ²)
X	9, 12, 15	—
Y	—	384

What is the perimeter, in meters, of plot Y ?

(A) 32

(B) 96 ✓

(C) 256

(D) 288

1. X is a right triangle (9-12-15): area = $\frac{1}{2}(9)(12) = 54$, perimeter = 36.

2. Area ratio = $\frac{384}{54} = \frac{64}{9}$, so the scale factor is $k = \sqrt{\frac{64}{9}} = \frac{8}{3}$.

3. Perimeter of $Y = 36 \cdot \frac{8}{3} = 96$.

Answer: (B)

23.

HARD · NO CALCULATOR NEEDED

Two sides of a triangle have lengths 8 and 10. The area of the triangle is 20. The angle between these two sides has a measure of C° , where $0 < C < 180$. What is the sum of all possible values of C ?

1. Area = $\frac{1}{2}(8)(10)\sin C = 40\sin C = 20 \Rightarrow \sin C = \frac{1}{2}$.

2. In $(0^\circ, 180^\circ)$, $\sin C = \frac{1}{2}$ has two solutions: $C = 30^\circ$ and $C = 150^\circ$.

3. Sum = $30 + 150 = 180$.

Answer: 180

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

In triangle ABC , $AB = 20$ and $BC = 12$. Point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that $\overline{DE} \parallel \overline{BC}$, and the area of triangle ADE equals the area of quadrilateral $DBCE$. What is the value of $(AD)(DE)$?

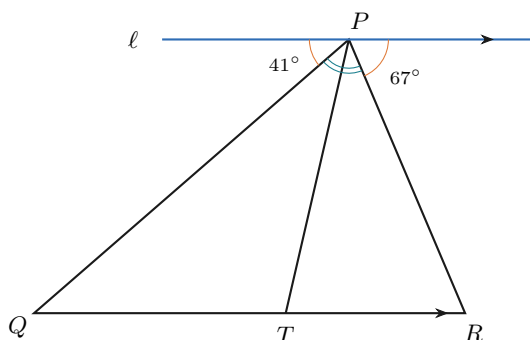
1. $\triangle ADE \sim \triangle ABC$ with ratio $k = \frac{AD}{AB}$. Equal areas means $[ADE] = \frac{1}{2}[ABC]$.
2. Areas scale by k^2 : $k^2 = \frac{1}{2}$, so $k = \frac{1}{\sqrt{2}}$.
3. $AD = 20k = \frac{20}{\sqrt{2}} = 10\sqrt{2}$ and $DE = 12k = \frac{12}{\sqrt{2}} = 6\sqrt{2}$.
4. $(AD)(DE) = (10\sqrt{2})(6\sqrt{2}) = 60 \cdot 2 = 120$.

Answer: 120

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the figure, line ℓ is parallel to side $\overline{QR}$ of triangle PQR . Point T lies on $\overline{QR}$, and $\overline{PT}$ bisects $\angle QPR$. What is the measure, in degrees, of $\angle PTR$?



(A) 72

(B) 36

(C) 103

(D) 77 ✓

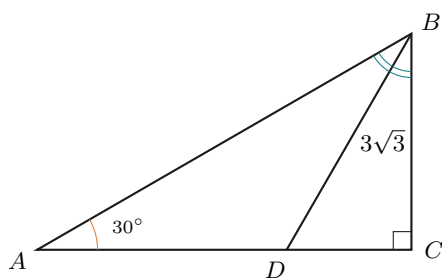
1. Alternate interior angles with $\ell \parallel \overline{QR}$: $\angle Q = 41^\circ$ and $\angle R = 67^\circ$.
2. Triangle sum: $\angle QPR = 180 - 41 - 67 = 72^\circ$; the bisector gives $\angle QPT = \angle TPR = 36^\circ$.
3. $\angle PTR$ is an exterior angle of $\triangle PQT$, so $\angle PTR = \angle Q + \angle QPT = 41 + 36 = 77$.
4. Check in $\triangle PTR$: $\angle PTR = 180 - 36 - 67 = 77$.

Answer: (D)

26.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the figure, $\angle C = 90^\circ$, $\angle A = 30^\circ$, $BC = 3\sqrt{3}$, and $\overline{BD}$ bisects $\angle ABC$. What is the length of $\overline{AD}$?



1. $\angle ABC = 90^\circ - 30^\circ = 60^\circ$, so $\angle DBC = 30^\circ$.
2. In $\triangle ABC$ (30° - 60° - 90°), BC is the short leg, so $AC = BC\sqrt{3} = 3\sqrt{3} \cdot \sqrt{3} = 9$.
3. In $\triangle BCD$ (30° - 60° - 90°), BC is the long leg, so $CD = \frac{BC}{\sqrt{3}} = 3$.
4. $AD = AC - CD = 9 - 3 = 6$. (Check: $\angle ABD = \angle A = 30^\circ$, so $AD = BD = 2CD = 6$.)

Answer: 6

27.

SAT CHALLENGE · NO CALCULATOR NEEDED

In right triangle ABC , angle C is the right angle. For a constant x , $\cos A = \frac{2x+4}{13}$ and $\sin B = \frac{5x-8}{13}$. What is the value of $\tan A$?

- (A) $\frac{5}{12}$ ✓ (B) $\frac{12}{5}$
 (C) $\frac{5}{13}$ (D) $\frac{12}{13}$

1. A and B are the acute angles of a right triangle, so they are complementary and $\cos A = \sin B$.
2. $2x + 4 = 5x - 8 \Rightarrow 3x = 12 \Rightarrow x = 4$, so $\cos A = \frac{12}{13}$.
3. Adjacent = 12, hypotenuse = 13, so opposite = $\sqrt{13^2 - 12^2} = 5$ (the triple 5-12-13).
4. $\tan A = \frac{\text{opp}}{\text{adj}} = \frac{5}{12}$.

Answer: (A)

28.

SAT CHALLENGE · NO CALCULATOR NEEDED

How many different triangles have integer side lengths and a longest side of exactly 10? (Triangles with the same three side lengths are counted once.)

1. Write the sides as $a \leq b \leq 10$. The triangle inequality (longest side $<$ sum of the others) requires $a + b > 10$.
2. For each b , the allowed a run from $11 - b$ to b , giving $2b - 10$ values: $b = 6, 7, 8, 9, 10$ give 2, 4, 6, 8, 10.
3. Total = $2 + 4 + 6 + 8 + 10 = 30$.

Answer: 30**29.**

SAT CHALLENGE · NO CALCULATOR NEEDED

Triangle Q is similar to triangle P , and the area of Q is 125% greater than the area of P . The sum of the lengths of an altitude of P and the corresponding altitude of Q is 15 centimeters. What is the length, in centimeters, of the altitude of Q ?

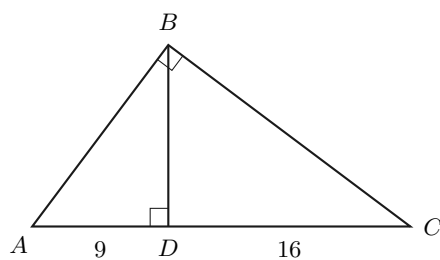
1. “125% greater” means the area of Q is 2.25 times the area of P .
2. Areas scale by k^2 : $k^2 = 2.25 \Rightarrow k = 1.5$, so $\text{altitude}_Q = 1.5 \cdot \text{altitude}_P$.
3. $\text{altitude}_P + 1.5 \text{altitude}_P = 15 \Rightarrow \text{altitude}_P = 6$.
4. $\text{altitude}_Q = 1.5 \cdot 6 = 9$.

Answer: 9

30.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the figure, $\angle ABC$ is a right angle and $\overline{BD}$ is perpendicular to $\overline{AC}$. What is the value of $\tan \angle ABD$?



(A) $\frac{9}{16}$

(B) $\frac{3}{4}$ ✓

(C) $\frac{4}{3}$

(D) $\frac{3}{5}$

1. In right $\triangle ABC$ with altitude $\overline{BD}$ to the hypotenuse, $BD^2 = AD \cdot DC = 9 \cdot 16 = 144$, so $BD = 12$.

2. In right $\triangle ABD$, the angle at B has opposite side $AD = 9$ and adjacent side $BD = 12$.

3. $\tan \angle ABD = \frac{9}{12} = \frac{3}{4}$. (Check: $\angle ABD = \angle C$, and $\tan C = \frac{AB}{BC} = \frac{15}{20} = \frac{3}{4}$.)

Answer: (B)

31.

SAT CHALLENGE · NO CALCULATOR NEEDED

In triangle ABC , the angle at C is a right angle. Point D lies on $\overline{AB}$ so that $\overline{CD} \perp \overline{AB}$. The area of triangle ACD is 48 and the area of triangle CDB is 108. What is the length of $\overline{AB}$?

1. Triangles ACD and CDB share the height CD , so their areas are in the ratio of their bases: $\frac{AD}{DB} = \frac{48}{108} = \frac{4}{9}$.

2. Let $AD = 4k$ and $DB = 9k$. In the right triangle ABC the altitude satisfies $CD^2 = AD \cdot DB = 36k^2$, so $CD = 6k$.

3. Area of ACD : $\frac{1}{2}(4k)(6k) = 12k^2 = 48 \Rightarrow k^2 = 4 \Rightarrow k = 2$.

4. $AB = AD + DB = 13k = 26$. (Check: $[CDB] = \frac{1}{2}(18)(12) = 108$.)

Answer: 26

32.

SAT CHALLENGE · NO CALCULATOR NEEDED

In triangle PQR , the angle at Q is a right angle. Point S lies on side $\overline{PR}$ so that $\overline{QS} \perp \overline{PR}$. If $\tan \angle QRP = \frac{4}{3}$ and $PS = 16$, what is the length of $\overline{SR}$?

(A) 15

(B) 25

(C) 9 ✓

(D) 12

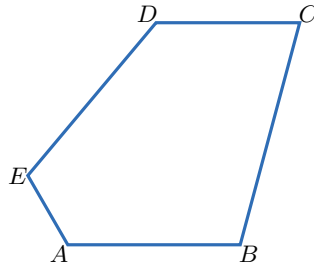
1. $\tan \angle QRP = \frac{PQ}{QR} = \frac{4}{3}$, so the legs are in the ratio 4 : 3 and the triangle is a 3-4-5 triangle: $PQ = 4k$, $QR = 3k$, $PR = 5k$.
2. Altitude relation $PQ^2 = PS \cdot PR$: $16k^2 = 16 \cdot 5k \Rightarrow k = 5$. So $PR = 25$.
3. $SR = PR - PS = 25 - 16 = 9$.
4. Check: $QR^2 = SR \cdot PR$ gives $15^2 = 9 \cdot 25$. The other values are $QS = 12$, $QR = 15$, $PR = 25$, which are different segments.

Answer: (C)

33.

SAT CHALLENGE · NO CALCULATOR NEEDED

In convex pentagon $ABCDE$ shown (not drawn to scale), side $\overline{AB}$ is parallel to side $\overline{DC}$. The measures of four interior angles are $\angle A = (6x)^\circ$, $\angle B = (5x + 5)^\circ$, $\angle C = (3x + 15)^\circ$ and $\angle D = (7x - 10)^\circ$. What is the measure, in degrees, of $\angle E$?



(A) 110 ✓

(B) 70

(C) 105

(D) 130

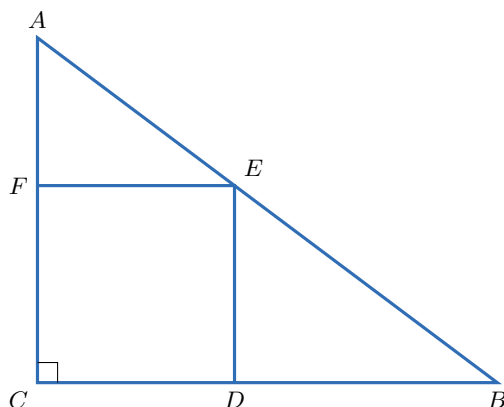
1. $\overline{AB} \parallel \overline{DC}$ with transversal $\overline{BC}$: $\angle B + \angle C = 180^\circ$ (co-interior).
2. $(5x + 5) + (3x + 15) = 180 \Rightarrow 8x = 160 \Rightarrow x = 20$.
3. $\angle A = 120^\circ$, $\angle B = 105^\circ$, $\angle C = 75^\circ$, $\angle D = 130^\circ$.
4. Pentagon angle sum $= (5 - 2) \cdot 180^\circ = 540^\circ$.
5. $\angle E = 540 - (120 + 105 + 75 + 130) = 110^\circ$.

Answer: (A)

34.

SAT CHALLENGE · NO CALCULATOR NEEDED

In right triangle ABC , $\angle C = 90^\circ$, $AB = 20$, and the area of the triangle is 96. Square $CDEF$ is inscribed in the triangle, with D on $\overline{BC}$, F on $\overline{AC}$, and E on $\overline{AB}$. What is the side length of the square?



(A) $\frac{24}{7}$

(B) $\frac{48}{5}$

(C) $\frac{96}{7}$

(D) $\frac{48}{7}$ ✓

1. Let the legs be $a = BC$, $b = AC$. Area: $\frac{1}{2}ab = 96 \Rightarrow ab = 192$. Pythagorean: $a^2 + b^2 = 400$.

2. $(a + b)^2 = a^2 + b^2 + 2ab = 400 + 384 = 784 \Rightarrow a + b = 28$ (no quadratic needed).

3. Triangle $FAE \sim$ triangle CAB (since $\overline{FE} \parallel \overline{CB}$): $\frac{AF}{AC} = \frac{FE}{CB}$, i.e. $\frac{b-s}{b} = \frac{s}{a}$.

4. This gives $s \left(\frac{1}{a} + \frac{1}{b} \right) = 1$, so $s = \frac{ab}{a+b} = \frac{192}{28} = \frac{48}{7}$.

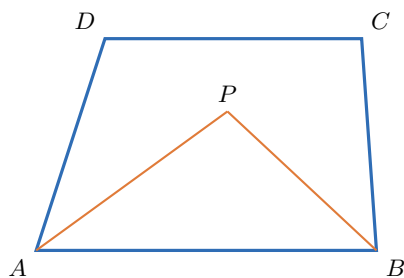
5. (Legs are 12 and 16; $\frac{12 \cdot 16}{28} = \frac{48}{7}$.)

Answer: (D)

35.

SAT CHALLENGE · NO CALCULATOR NEEDED

In trapezoid $ABCD$ shown (not drawn to scale), $\overline{AB} \parallel \overline{DC}$, $\angle D = 108^\circ$, and $\angle C = 94^\circ$. The bisector of $\angle DAB$ and the bisector of $\angle ABC$ intersect at point P inside the trapezoid. What is the measure, in degrees, of the obtuse angle formed at P by these two bisectors?



(A) 22

(B) 108

(C) 101 ✓

(D) 79

1. $\overline{AB} \parallel \overline{DC}$: $\angle A + \angle D = 180^\circ \Rightarrow \angle A = 72^\circ$; $\angle B + \angle C = 180^\circ \Rightarrow \angle B = 86^\circ$.
2. Half-angles: $\angle PAB = 36^\circ$, $\angle PBA = 43^\circ$.
3. In triangle APB : $\angle APB = 180 - 36 - 43 = 101^\circ$, which is obtuse.
4. The vertical/supplementary angle is 79° (acute), so the obtuse angle is 101° .

Answer: (C)

36.

SAT CHALLENGE · NO CALCULATOR NEEDED

In right triangle PQR , $\angle R = 90^\circ$ and $\sin P + \sin Q = \frac{\sqrt{6}}{2}$. What is the value of $\tan P + \tan Q$?

1. $\angle Q = 90^\circ - \angle P$, so $\sin Q = \cos P$ and $\tan Q = \frac{1}{\tan P}$.
2. Square the given sum: $\sin^2 P + 2 \sin P \cos P + \cos^2 P = \frac{6}{4} \Rightarrow 1 + 2 \sin P \cos P = \frac{3}{2}$.
3. $\sin P \cos P = \frac{1}{4}$.
4. $\tan P + \tan Q = \frac{\sin P}{\cos P} + \frac{\cos P}{\sin P} = \frac{1}{\sin P \cos P} = 4$.

Answer: 4

37.

SAT CHALLENGE · NO CALCULATOR NEEDED

Two lines intersect at point O , and the smallest of the four angles they form measures 50° . A ray from O bisects each of the four angles. Together with the two lines, this gives eight rays from O , which divide the full turn into eight small angles. A *span* is an angle formed by 1, 2 or 3 consecutive small angles. Which of the following is NOT the measure of any span?

(A) 155° (B) 140° ✓(C) 115° (D) 130°

- The lines cut the plane into angles of 50° and 130° . Bisecting gives pieces of 25° and 65° .
- Going around O the small angles are 25, 25, 65, 65, 25, 25, 65, 65 (degrees), repeating.
- One small angle: 25 or 65. Two consecutive: $25+25 = 50$, $25+65 = 90$, $65+65 = 130$. Three consecutive: $25+25+65 = 115$, $25+65+65 = 155$ (and the same sums in the other order).
- The possible spans are 25, 65, 50, 90, 130, 115, 155. The values 115, 130 and 155 occur, but 140 does not.

Answer: (B)

38.

SAT CHALLENGE · CALCULATOR

In right triangle ABC , the right angle is at C and $AB = 10$. Point D lies on $\overline{AB}$ with $\overline{CD} \perp \overline{AB}$ and $CD = 4$. If $AD > DB$, what is the value of $\cos A$?

(A) $\frac{\sqrt{5}}{5}$ (B) $\frac{4}{5}$ (C) $\frac{1}{2}$ (D) $\frac{2\sqrt{5}}{5}$ ✓

- Let $AD = p$ and $DB = q$. Then $p + q = 10$, and the altitude to the hypotenuse gives $CD^2 = pq = 16$.
- $p(10 - p) = 16 \Rightarrow p^2 - 10p + 16 = 0 \Rightarrow (p - 2)(p - 8) = 0$. Since $AD > DB$, $AD = 8$ and $DB = 2$.
- In right triangle ACD : $AC = \sqrt{8^2 + 4^2} = \sqrt{80} = 4\sqrt{5}$.
- $\cos A = \frac{AD}{AC} = \frac{8}{4\sqrt{5}} = \frac{2}{\sqrt{5}} = \frac{2\sqrt{5}}{5}$.
- (Other choices: $\frac{\sqrt{5}}{5} = \sin A$, $\frac{4}{5} = \frac{AD}{AB}$, $\frac{1}{2} = \tan A$.)

Answer: (D)

39.

SAT CHALLENGE · NO CALCULATOR NEEDED

In equilateral triangle ABC , point D lies on side $\overline{BC}$ such that $BD = \frac{1}{4}BC$. What is the value of $\tan \angle DAC$?

- (A) $\frac{3\sqrt{3}}{5}$ ✓ (B) $\frac{\sqrt{3}}{7}$
 (C) $\frac{5\sqrt{3}}{9}$ (D) $\frac{3\sqrt{3}}{7}$

- Place $B = (0, 0)$ and $C = (4, 0)$, so $A = (2, 2\sqrt{3})$ and $D = (1, 0)$.
- Vectors from A : $\vec{AC} = (2, -2\sqrt{3})$ and $\vec{AD} = (-1, -2\sqrt{3})$.
- Cross product (magnitude): $|2(-2\sqrt{3}) - (-2\sqrt{3})(-1)| = |-4\sqrt{3} - 2\sqrt{3}| = 6\sqrt{3}$.
- Dot product: $2(-1) + (-2\sqrt{3})(-2\sqrt{3}) = -2 + 12 = 10$.
- $\tan \angle DAC = \frac{6\sqrt{3}}{10} = \frac{3\sqrt{3}}{5}$.
- (Check: $\tan \angle BAD = \frac{\sqrt{3}}{7}$ by the same method, and $\frac{\sqrt{3}/7 + 3\sqrt{3}/5}{1 - \frac{9}{35}} = \sqrt{3} = \tan 60^\circ$.)

Answer: (A)

40.

SAT CHALLENGE · NO CALCULATOR NEEDED

In right triangle ABC , the right angle is at C and $\tan A = \frac{5}{12}$. The area of the triangle is 120. Point D lies on $\overline{AB}$ such that $\overline{CD} \perp \overline{AB}$. What is the length of $\overline{AD}$?

- (A) $\frac{50}{13}$ (B) $\frac{288}{13}$ ✓
 (C) $\frac{120}{13}$ (D) 24

- $\tan A = \frac{BC}{AC} = \frac{5}{12}$, so let $BC = 5k$ and $AC = 12k$. Then $AB = 13k$.
- Area: $\frac{1}{2}(12k)(5k) = 30k^2 = 120 \Rightarrow k = 2$. So $AC = 24$, $BC = 10$, $AB = 26$.
- In a right triangle, each leg squared equals the hypotenuse times the adjacent segment: $AC^2 = AD \cdot AB$.
- $AD = \frac{24^2}{26} = \frac{576}{26} = \frac{288}{13}$.
- (Other options: $\frac{50}{13} = DB$, $\frac{120}{13} = CD$, and $24 = AC$.)

Answer: (B)

41.

SAT CHALLENGE · NO CALCULATOR NEEDED

In triangle ABC , point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that $\overline{DE}$ is parallel to $\overline{BC}$. The measure of $\angle ADE$ is $(4x - 10)^\circ$, the measure of $\angle ABC$ is $(2x + 30)^\circ$, and the measure of $\angle ACB$ is $(x + 25)^\circ$. What is the measure, in degrees, of $\angle BAC$?

(A) 115

(B) 45

(C) 65 ✓

(D) 70

1. Since $\overline{DE} \parallel \overline{BC}$, with transversal $\overline{AB}$, the corresponding angles are equal: $\angle ADE = \angle ABC$.

2. $4x - 10 = 2x + 30 \Rightarrow x = 20$.

3. Then $\angle ABC = 70^\circ$ and $\angle ACB = 20 + 25 = 45^\circ$.

4. Angle sum in triangle ABC : $\angle BAC = 180 - 70 - 45 = 65^\circ$.

5. (Pairing $\angle ADE$ with $\angle ACB$ instead would give $x = \frac{35}{3}$ and a wrong triangle.)

Answer: (C)

42.

SAT CHALLENGE · CALCULATOR

For $0 < x < 45$, the equation

$$\sin((5x - 20)^\circ) = \cos((x + 14)^\circ)$$

has more than one solution. What is the sum of all solutions x in this interval?

1. Write the cosine as a sine: $\cos((x + 14)^\circ) = \sin((76 - x)^\circ)$.

2. Equal sines give two branches. Branch 1: $5x - 20 = 76 - x + 360n \Rightarrow 6x = 96 + 360n$. Only $n = 0$ lies in $(0, 45)$: $x = 16$.

3. Branch 2: $5x - 20 = 180 - (76 - x) + 360n = 104 + x + 360n \Rightarrow 4x = 124 + 360n$. Only $n = 0$ lies in $(0, 45)$: $x = 31$.

4. Check: $x = 16$ gives $\sin 60^\circ = \cos 30^\circ = \frac{\sqrt{3}}{2}$; $x = 31$ gives $\sin 135^\circ = \cos 45^\circ = \frac{\sqrt{2}}{2}$. Both satisfy the equation.

5. The sum is $16 + 31 = 47$.

Answer: 47

43.

SAT CHALLENGE · NO CALCULATOR NEEDED

Points A , B , C and D are consecutive vertices of a regular polygon with n sides, and $\overline{AC}$ is a diagonal. The measure of $\angle BAC$ is $(x + 3)^\circ$ and the measure of $\angle ACD$ is $(10x + 15)^\circ$, where x is a constant. What is the value of n ?

(A) 12 ✓

(B) 15

(C) 24

(D) 8

- Triangle ABC is isosceles with $AB = BC$ and apex angle $\angle ABC$ equal to the interior angle $180^\circ - \frac{360^\circ}{n}$. So $\angle BAC = \angle BCA = \frac{180^\circ}{n}$.
- The interior angle at C is $\angle BCD = 180^\circ - \frac{360^\circ}{n}$, and it is split by the diagonal: $\angle ACD = \angle BCD - \angle BCA = 180^\circ - \frac{540^\circ}{n}$.
- Equate with the given expressions: $x + 3 = \frac{180}{n}$ and $10x + 15 = 180 - \frac{540}{n}$.
- Substitute $x = \frac{180}{n} - 3$: $\frac{1800}{n} - 30 + 15 = 180 - \frac{540}{n} \Rightarrow \frac{2340}{n} = 195 \Rightarrow n = 12$.
- Check: $\angle BAC = 15^\circ$ gives $x = 12$; $\angle ACD = 10(12) + 15 = 135^\circ = 180^\circ - \frac{540^\circ}{12}$.
- $n = 12$. (Treating 135° as the interior angle gives 8; treating 15° as the exterior angle gives 24.)

Answer: (A)

44.

SAT CHALLENGE · NO CALCULATOR NEEDED

In right triangle ABC , the angle at C is a right angle. Point D lies on $\overline{AB}$ so that $\overline{CD} \perp \overline{AB}$. The length of $\overline{CD}$ is 6, and DB is 5 units longer than AD . What is the value of $\sin B$?

(A) $\frac{3\sqrt{13}}{13}$ (B) $\frac{4}{13}$ (C) $\frac{6}{13}$ (D) $\frac{2\sqrt{13}}{13}$ ✓

- Let $AD = t$, so $DB = t + 5$. The altitude to the hypotenuse satisfies $CD^2 = AD \cdot DB$: $36 = t(t + 5)$.
- $t^2 + 5t - 36 = 0 \Rightarrow (t + 9)(t - 4) = 0$, so $t = 4$ (lengths are positive). Then $AD = 4$, $DB = 9$, $AB = 13$.
- Leg relation: $BC^2 = DB \cdot AB = 9 \cdot 13 = 117$, so $BC = 3\sqrt{13}$.
- $\sin B = \frac{AC}{AB}$ or, in triangle CDB , $\frac{CD}{CB} = \frac{6}{3\sqrt{13}} = \frac{2}{\sqrt{13}} = \frac{2\sqrt{13}}{13}$.
- (Check: $AC^2 = AD \cdot AB = 52$, $AC = 2\sqrt{13}$, and $\frac{2\sqrt{13}}{13}$ again. The option $\frac{3\sqrt{13}}{13}$ is $\sin A$, $\frac{4}{13}$ is $\cos^2 A$ and $\frac{6}{13}$ is $\frac{CD}{AB}$.)

Answer: (D)

45.

SAT CHALLENGE · NO CALCULATOR NEEDED

A convex polygon has n sides. Its interior angles, listed from smallest to largest, form an arithmetic sequence with smallest angle 120° and common difference 5° . How many sides does the polygon have?

(A) 18

(B) 7

(C) 9 ✓

(D) 16

1. Sum of the angles as an arithmetic series: $\frac{n}{2}(2 \cdot 120 + (n - 1) \cdot 5) = \frac{n}{2}(235 + 5n)$.
2. Angle sum of an n -gon: $180(n - 2)$. Equate: $n(235 + 5n) = 360(n - 2) \Rightarrow 5n^2 - 125n + 720 = 0 \Rightarrow n^2 - 25n + 144 = 0$.
3. Factor: $(n - 9)(n - 16) = 0$, so $n = 9$ or $n = 16$.
4. Convexity needs every angle below 180° . Largest angle: $120 + 5(n - 1)$. For $n = 16$ it is $195^\circ > 180^\circ$, so reject.
5. For $n = 9$ the largest angle is $160^\circ < 180^\circ$ and the sum is $\frac{9}{2}(240 + 40) = 1260 = 180 \cdot 7$. The polygon has 9 sides.

Answer: (C)

46.

SAT CHALLENGE · NO CALCULATOR NEEDED

Two lines intersect and form four angles. The acute angles each measure $(2k + 4)^\circ$, where k is a constant. Each obtuse angle measures 4 degrees more than 3 times the measure of an acute angle. Which of the following CANNOT be the sum, in degrees, of the measures of two of the four angles?

(A) 180

(B) 224 ✓

(C) 272

(D) 88

1. An acute angle and an obtuse angle at an intersection are supplementary: $(2k + 4) + [3(2k + 4) + 4] = 180$.
2. $4(2k + 4) + 4 = 180 \Rightarrow 8k + 20 = 180 \Rightarrow k = 20$.
3. Acute angles: $2(20) + 4 = 44^\circ$. Obtuse angles: $3(44) + 4 = 136^\circ$. Check: $44 + 136 = 180$.
4. Sums of two of the four angles: acute + acute = 88; acute + obtuse = 180; obtuse + obtuse = 272.
5. 224 is not among them, so it cannot be a sum.

Answer: (B)

47.

SAT CHALLENGE · NO CALCULATOR NEEDED

In triangle ABC , the measure of angle A is $(3x + 11)^\circ$, the measure of angle B is $(2x + 4)^\circ$, and the measure of angle C is 30° . What is the value of

$$(\sin A \cos B + \cos A \sin B)(\cos A \cos B - \sin A \sin B)?$$

(A) $\frac{\sqrt{3}}{4}$

(B) $-\frac{1}{4}$

(C) $\frac{1}{4}$

(D) $-\frac{\sqrt{3}}{4}$ ✓

1. The angles of a triangle sum to 180° , so $A + B = 180^\circ - 30^\circ = 150^\circ$. (Solving, $5x + 15 = 150$, $x = 27$, but the value of x is not needed.)
2. The first factor is $\sin(A + B) = \sin 150^\circ = \frac{1}{2}$.
3. The second factor is $\cos(A + B) = \cos 150^\circ = -\frac{\sqrt{3}}{2}$.
4. Product: $\frac{1}{2} \cdot \left(-\frac{\sqrt{3}}{2}\right) = -\frac{\sqrt{3}}{4}$. (Using $\sin 30^\circ \cos 30^\circ$ instead would give the wrong sign.)

Answer: (D)

48.

SAT CHALLENGE · NO CALCULATOR NEEDED

In triangle PQR , the measure of angle P is 30° and $PR = 14$. The length $QR = a$ is a positive integer with $a \leq 20$. For how many values of a does the given information determine exactly one triangle, up to congruence?

1. Here the known angle is $P = 30^\circ$, the adjacent side is $PR = 14$ and the opposite side is $QR = a$ (an SSA situation). The height is $14 \sin 30^\circ = 7$.
2. $a < 7$: side QR is too short to reach line PQ , no triangle. $a = 7$: it just touches, one right triangle.
3. $7 < a < 14$: two different triangles (two positions for Q). $a \geq 14$: only one position, since the second intersection lies on the opposite side of P .
4. Exactly one triangle: $a = 7$ and $a = 14, 15, \dots, 20$, which is $1 + 7 = 8$ values.

Answer: 8